

INDUSTRIAL PROGRAMMING LANGUAGE

MDA 3044

LAB 7 & 8

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LAB 7

QUESTION 2

```java
package lab7;
import java.util.Scanner;

public class Q2 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // User input vector size
        System.out.print("Enter vector size (n): ");
        int n = scanner.nextInt();
        scanner.nextLine();

        System.out.println("Enter elements for vector 1 (space separated):");
        double[] v1 = vector(scanner, n);

        System.out.println("Enter elements for vector 2 (space separated):");
        double[] v2 = vector(scanner, n);

        // Calculate and display result
        System.out.println("Dot product: " + computeDotProduct(v1, v2));
        scanner.close();
    }

    public static double[] vector(Scanner scanner, int n) {
        String input = scanner.nextLine().trim();
        String[] elements = input.split("\\s+");

        // Validate input size
        if (elements.length != n) {
            System.err.println("Error: Does not match the predefined size: " + n + " elements.");
            System.exit(1);
        }

        // Parse elements
        double[] vector = new double[n];
        for (int i = 0; i < n; i++) {
            try {
                vector[i] = Double.parseDouble(elements[i]);
            } catch (NumberFormatException e) {
                System.err.println("Error: Invalid number format at position " + (i+1));
                System.exit(1);
            }
        }
        return vector;
    }

    public static double computeDotProduct(double[] v1, double[] v2) {
        double sum = 0;
        for (int i = 0; i < v1.length; i++) {
            sum += v1[i] * v2[i];
        }
        return sum;
    }
}
```

```
Enter vector size (n): 4
Enter elements for vector 1 (space separated):
2 3 4 5
Enter elements for vector 2 (space separated):
1 2 3 4
Dot product: 40.0
```

QUESTION 3

```java
package lab7;
import java.util.Scanner;

public class Q3{

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = 10;  // Number of integers to read

        int[] numbers = new int[n];

        System.out.println("Enter " + n + " integers");

        // Read integers
        for (int i = 0; i < n; i++) {
            System.out.print("Number " + (i + 1) + ": ");
            numbers[i] = scanner.nextInt();
        }

        // Display in reverse order
        System.out.println("Numbers in reverse order:");
        printReverse(numbers);

        scanner.close();
    }

    // Method to print array elements in reverse order
    public static void printReverse(int[] arr) {
        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
}
```

```
Enter 10 integers
Number 1: 1
Number 2: 2
Number 3: 3
Number 4: 4
Number 5: 5
Number 6: 6
Number 7: 7
Number 8: 8
Number 9: 9
Number 10: 10
Numbers in reverse order:
10 9 8 7 6 5 4 3 2 1
```

LAB 8

QUESTION 1

```java
package lab8;
import java.util.Scanner;

public class Q1 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        double[][] matrix = new double[4][4];

        System.out.println("Enter a 4x4 matrix (16 numbers):");
        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 4; j++) {
                matrix[i][j] = scanner.nextDouble();
            }
        }

        double sum = sumMajorDiagonal(matrix);
        System.out.printf("Sum of the major diagonal is: %.2f\n", sum);

        scanner.close();
    }

    // Method to sum the major diagonal of an n x n matrix
    public static double sumMajorDiagonal(double[][] m) {
        double sum = 0;
        int n = m.length;  // assuming square matrix
        for (int i = 0; i < n; i++) {
            sum += m[i][i];
        }
        return sum;
    }
}
```

```
Enter a 4x4 matrix (16 numbers):
1 2 3 4
2 3 5 4
5 4 3 2
1 2 3 4
Sum of the major diagonal is: 11.00
```

QUESTION 2

```java
package lab8;
import java.util.Scanner;

public class Q2 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        double[][] matrixA = new double[3][3];
        double[][] matrixB = new double[3][3];

        System.out.println("Enter matrix A (3x3):");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrixA[i][j] = scanner.nextDouble();
            }
        }

        System.out.println("Enter matrix B (3x3):");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrixB[i][j] = scanner.nextDouble();
            }
        }

        double[][] sumMatrix = addMatrix(matrixA, matrixB);

        System.out.println("Sum of matrix A and B is:");
        printMatrix(sumMatrix);

        scanner.close();
    }

    // Method to add two matrices
    public static double[][] addMatrix(double[][] a, double[][] b) {
        int rows = a.length;
        int cols = a[0].length;

        double[][] c = new double[rows][cols];

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                c[i][j] = a[i][j] + b[i][j];
            }
        }
        return c;
    }

    // Helper method to print a matrix
    public static void printMatrix(double[][] matrix) {
        for (double[] row : matrix) {
            for (double val : row) {
                System.out.printf("%.2f ", val);
            }
            System.out.println();
        }
    }
}
```

```
Enter matrix A (3x3):

1 2 3
3 4 5
5 6 7
Enter matrix B (3x3):
9 8 7
7 6 5
5 4 3
Sum of matrix A and B is:
10.00 10.00 10.00
10.00 10.00 10.00
10.00 10.00 10.00
```